Model selection and performance estimation can diverge

Road-sensing evaluation audit | Three sources, different grouping evidence

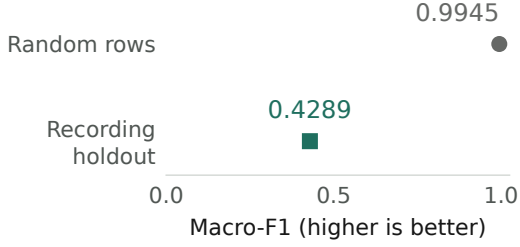
RoadSens: 101 CSV sources
Filename groups; not verified roads

PVS: 9 acquisitions
Vehicle and driver are confounded

Mexico: 2 route families
Documentation-derived groups

1 Audit the evaluation unit

RoadSens primary policy, RF100



Different prediction targets;
the score gap alone is not causal evidence.

2 Separate the endpoints

PVS primary grid: random to matched selection

Selection: scenario outer macro-F1

0.7640 → 0.8359

Estimation: mean absolute error (lower is better)

Vehicle/driver	0.0530 → 0.0445
Scenario	0.1696 → 0.1357

Required: MAE reduction ≥ 0.02 on both axes.

Declared two-axis criterion: NOT MET

3 Retain the contrary evidence

Disjoint windows and post hoc common-budget diagnostics: MAE worsens on both PVS axes.

Mexico released subset: both selectors choose the same RF; selection advantage is zero.

Bounded evidence, not a universal validation improvement. Same PVS acquisitions; no population intervals.